

B105 Applied Statistical Modelling

Statistical Analysis of Online Food Delivery Adoption

Student Name: Sumit Kumar

Student ID: GH1036298

Table of Contents

1. Introduction.....	4
1.1 Background.....	4
1.2 Business Problem.....	4
1.3 Business Questions	4
1.4 Hypotheses.....	5
2. Dataset and Data Preparation.....	5
2.1 Dataset.....	5
2.2 Initial Data Inspection.....	6
3. Data Cleaning and Pre-processing.....	7
3.1 Duplicate Observations.....	7
3.2 Text Standardization	8
4. Sampling and Final Analytical Dataset.....	9
5. Descriptive Statistics and Exploratory Data Analysis	9
5.1 Numerical Descriptive Statistics.....	9
6. Categorical Descriptive Analysis.....	10
7. Exploratory Data Analysis.....	12
8. Inferential Statistical Analysis	15
8.1 Statistical Method	15
9. Business Question 1: Customer Type and Output.....	16
10. Business Question 2: Monthly Income and Output	18
11. Business Question 3: Occupation and Output	19
12. Chi-Square Assumption Validation	20
13. Monte Carlo Robustness Check.....	21
14. Overall Findings and Business Implications.....	22

15. Limitations	22
16. Recommendations and Future Work	23
17. Conclusion	24
References.....	25

1. Introduction

1.1 Background

Online food delivery has become an important part of the modern food-service market. Customers can order meals online without having to physically visit restaurants, and convenience and accessibility are all important for consumers. Food delivery companies can use the data to help them understand what customer attributes are correlated with adoption to help with customer segmentation, marketing decisions and service planning.

This project examines an Online Food Delivery Dataset, looking to see if there is an association between whether the customer uses Online Food Delivery or not and customer characteristics. The analysis is targeted at features of customers such as customer type, monthly income and occupation. These variables offer various approaches to customers' behavior and may be useful in determining if there are differences in customers' adoption across customer types.

The data was analyzed using statistical packages like R and was completed with data inspection, data clean-up, descriptive analysis, exploratory data analysis (EDA), cross-tabulation and Pearson's chi-square tests. The strength of the association identified statistically was also measured using Cramer's V. The standard chi square tests also left some warnings related to approximation of the test, so a Monte Carlo chi square procedure was also performed as a robustness check.

1.2 Business Problem

The online food delivery business has to face the issue that whether people's characteristics correlate with the acceptance of the food delivery. Differences, if any, between customer categories could be misinterpreted as significant business trends without statistical evidence.

Therefore, the analysis investigates whether Customer Type, Monthly Income and Occupation are associated with the Output variable, where Output represents online food delivery adoption.

1.3 Business Questions

The analysis addresses three business questions:

- **BQ1:** Is there an association between Customer Type and online food delivery adoption?

- **BQ2:** Is there an association between Monthly Income and online food delivery adoption?
- **BQ3:** Is there an association between Occupation and online food delivery adoption?

1.4 Hypotheses

For BQ1:

- **H₀:** There is no association between Customer Type and online food delivery adoption.
- **H₁:** There is an association between Customer Type and online food delivery adoption.

For BQ2:

- **H₀:** There is no association between Monthly Income and online food delivery adoption.
- **H₁:** There is an association between Monthly Income and online food delivery adoption.

For BQ3:

- **H₀:** There is no association between Occupation and online food delivery adoption.
- **H₁:** There is an association between Occupation and online food delivery adoption.

A significance level of $\alpha = 0.05$ was used, corresponding to a **95% confidence level**.

```
>
> cat("Significance level (alpha):", alpha, "\n")
significance level (alpha): 0.05
> cat("Confidence level:", 1 - alpha, "\n")
Confidence level: 0.95
```

Figure 1 Significance and confidence levels used in the statistical analysis

2. Dataset and Data Preparation

2.1 Dataset

The chosen data set is an online food delivery dataset that includes demographic, socioeconomic and customer related data (Chotigo and Kadono, 2021). There were 14 variables and 388 observations in the original dataset.

The variables used are Age, Gender, Marital Status, Occupation, Monthly Income, Educational Qualifications, Family Size, Customer Type, latitude, longitude, Pin Code, Output and Feedback. There was also an X column in the data set.

This dataset was chosen due to its categorical variables that are suitable for studying associations with online food delivery adoption (Ahmad, 2023; Francioni et al., 2022). No transformation of the data was required when it was analysed because the problem investigated was statistical rather than a predictive modelling problem.

Previous studies have examined consumer intention and continued use of online food delivery services, highlighting the relevance of customer characteristics when studying adoption behavior (Hong et al., 2021; Chotigo and Kadono, 2021).

```
> # =====
> # B105 - LOAD ONLINE FOOD DELIVERY DATASET
> # =====
>
> data_raw <- read.csv(
+   "C:/Users/hp/Desktop/B105_Statistical_Modelling/online food delivery dataset.csv",
+   stringsAsFactors = FALSE
+ )
>
> # Check dataset
> dim(data_raw)
[1] 388 14
>
> head(data_raw)
  Age Gender Marital.Status Occupation Monthly.Income Educational.Qualifications
1  20 Female      Single      Student      No Income      Post Graduate
2  24 Female      Single      Student Below Rs.10000      Graduate
3  22 Male       Single      Student Below Rs.10000      Post Graduate
4  22 Female      Single      Student      No Income      Graduate
5  22 Male       Single      Student Below Rs.10000      Post Graduate
6  27 Female      Married Employee More than 50000      Post Graduate
 Family.size Customer.Type latitude longitude Pin.code Output Feedback X
1         4      Frequent  12.9766   77.5993  560001   Yes Positive Yes
2         3         Regular  12.9770   77.5773  560009   Yes Positive Yes
3         3         Regular  12.9551   77.6593  560017   Yes Negative Yes
4         6      Frequent  12.9473   77.5616  560019   Yes Positive Yes
5         4      Frequent  12.9850   77.5533  560010   Yes Positive Yes
6         2         Regular  12.9299   77.6848  560103   Yes Positive Yes
>

> str(data_raw)
'data.frame':   388 obs. of  14 variables:
 $ Age          : int  20 24 22 22 22 27 22 24 23 23 ...
 $ Gender       : chr  "Female" "Female" "Male" "Female" ...
 $ Marital.Status : chr  "Single" "Single" "Single" "Single" ...
 $ Occupation   : chr  "Student" "Student" "Student" "Student" ...
 $ Monthly.Income : chr  "No Income" "Below Rs.10000" "Below Rs.10000" "No I
 ncome" ...
 $ Educational.Qualifications: chr  "Post Graduate" "Graduate" "Post Graduate" "Graduat
 e" ...
 $ Family.size  : int  4 3 3 6 4 2 3 3 2 4 ...
 $ Customer.Type : chr  "Frequent" "Regular" "Regular" "Regular" "Frequent" ...
 $ latitude     : num  13.13 13.13 12.9 13 ...
 $ longitude    : num  77.6 77.6 77.7 77.6 77.6 ...
 $ Pin.code     : int  560001 560009 560017 560019 560010 560103 560009 56
 0042 560001 560048 ...
 $ Output       : chr  "Yes" "Yes" "Yes" "Yes" ...
 $ Feedback     : chr  "Positive" "Positive" "Negative" "Positive" ...
 $ X            : chr  "Yes" "Yes" "Yes" "Yes" ...
> # =====
```

Figure 2 showing the dimensions, first observations and structure of the raw dataset

2.2 Initial Data Inspection

The first check was of variable names, dimensions, structure, summary statistics, and missing values of the datasets. There were 14 columns and 388 rows in the data set. Variables were examined prior to cleaning with the `summary(data_raw)` command. The presence of missing values was checked and no missing values were found in the original data.

The number and percentage of missing values were obtained for each variable and thus data quality was evaluated prior to analysis.

3. Data Cleaning and Pre-processing

3.1 Duplicate Observations

`Duplicated(data_raw)` was used to examine duplicate observations. The data set had 103 duplicated rows or about 26.55% of data points.

If one considers the frequency-based statistical analysis of the data, the presence of a large number of exact duplicate observations may have an effect if they were repeated copies of observations instead of independent observations. Thus, exact replications were eliminated prior to conducting the inferential analysis.

In addition, there was a column named X that was the same size as the Output column and held the same Output information. The following line was used to remove this redundant variable:

```
data_clean <- data_raw %>%  
  select(-X)
```

After removing the redundant variable, exact duplicate rows were removed using `distinct()`.

```

> # =====
> # 3. DUPLICATE CHECK
> # =====
>
> # Number of duplicated rows
> sum(duplicated(data_raw))
[1] 103
>
> # Percentage of duplicated rows
> round(
+   sum(duplicated(data_raw)) / nrow(data_raw) * 100,
+   2
+ )
[1] 26.55
>
> # Display duplicated observations
> data_raw[duplicated(data_raw), ]
  Age Gender Marital.Status Occupation Monthly.Income
44  22 Female          Single      Student      No Income
49  23 Female          Single      Student      No Income
50  22 Female          Single      Student 10001 to 25000
53  24 Male           Single      Student      No Income
61  23 Male           Single      Student      No Income
69  23 Female          Single      Student      No Income
78  21 Male           Single      Student      No Income
87  24 Male           Single      Student      No Income
89  25 Male           Single      Student      No Income
94  24 Female          Single      Student      No Income
96  25 Male           Single      Student      No Income
97  23 Female          Single      Student      No Income
98  23 Female          Single      Student      No Income
101 24 Female          Single      Student 10001 to 25000
112 25 Male           Single      Student      No Income

```

Figure 3 Duplicate observation check in the raw dataset

3.2 Text Standardization

Categorical variables were checked for inconsistent whitespace. Gender, Marital Status, Occupation, Monthly Income, Educational Qualifications, Customer Type, Output and Feedback were the attributes that were trimmed by the trimws() function.

This was significant because there is a trailing space in the negative category of the original Feedback variable. Apparently the identical categories may be considered as separate categories without any standardisation.

Unique() and table() were used to verify the categorical variables after cleaning. The final cleaned data set consisted of 285 observations and 13 variables, and had no missing values or duplicate rows of data.

```

> # =====
> # 11. FINAL DATASET VERIFICATION
> # =====
>
> cat("Final sample size:", nrow(analysis_data), "\n")
Final sample size: 285
> cat("Final number of variables:", ncol(analysis_data), "\n")
Final number of variables: 13
> cat("Missing values:", sum(is.na(analysis_data)), "\n")
Missing values: 0
> cat("Duplicate rows:", sum(duplicated(analysis_data)), "\n")
Duplicate rows: 0
>
> cat("\nOutput distribution:\n")

```

Figure 4 Final analytical dataset verification after data cleaning

4. Sampling and Final Analytical Dataset

After cleaning, all valid unique observations were retained for the analysis:

```
analysis_data <- data_clean
```

Therefore, no additional random sampling was performed. The data analysis was the complete data set of the valid unique observations and not a smaller sample set.

The number of observations and variables in the final analytical data set was 285 observations and 13 variables. This was done so that all the observations that were valid and acceptable were used in the statistical analysis.

The cleaned dataset was saved as:

```
cleaned_online_food_delivery_data.csv
```

This also provides a reproducible version of the dataset used in the analysis.

5. Descriptive Statistics and Exploratory Data Analysis

5.1 Numerical Descriptive Statistics

Descriptive statistics were calculated for Age and Family Size.

Age

The average age of the customers was 24.68 years, and the median was 24 years. The ages ranged from 18-33 years and were approximately 3.04 years standard deviation.

The difference in the mean and median is relatively small so the centre of the age distribution is reasonably near, but the histogram also gives us some information.

Family Size

The mean family size was 3.27, and the median was 3. The average value of the standard deviation was about 1.36 and the number of family members ranged from 1 to 6.

The statistics give a high-level overview of the customer base prior to analyzing the relationships relevant to the business questions.

```
> # =====
> # STEP 12: DESCRIPTIVE STATISTICS AND EXPLORATORY DATA ANALYSIS
> # =====
> # -----
> # 12.1 Create folders for figures and outputs
> # -----
>
> if(!dir.exists("figures")) dir.create("figures")
> if(!dir.exists("outputs")) dir.create("outputs")
> # -----
> # 12.2 Descriptive statistics for numerical variables
> # -----
>
> # Age
> age_stats <- data.frame(
+   Mean = mean(analysis_data$Age, na.rm = TRUE),
+   Median = median(analysis_data$Age, na.rm = TRUE),
+   SD = sd(analysis_data$Age, na.rm = TRUE),
+   Minimum = min(analysis_data$Age, na.rm = TRUE),
+   Maximum = max(analysis_data$Age, na.rm = TRUE),
+ )
>
> age_stats
  Mean Median      SD Minimum Maximum
1 24.67719    24 3.040977     18      33
>
>
> # Family Size
> family_stats <- data.frame(
+   Mean = mean(analysis_data$Family.size, na.rm = TRUE),
+   Median = median(analysis_data$Family.size, na.rm = TRUE),
+   SD = sd(analysis_data$Family.size, na.rm = TRUE),
+   Minimum = min(analysis_data$Family.size, na.rm = TRUE),
+ )
>
> family_stats
  Mean Median      SD Minimum Maximum
1 3.270175     3 1.361178      1      6
>
>
> # Overall summary
> summary(analysis_data[, c("Age", "Family.size")])
   Age      Family.size
Min.  :18.00  Min.    :1.00
1st Qu.:23.00 1st Qu.:2.00
Median :24.00 Median :3.00
Mean   :24.68 Mean   :3.27
3rd Qu.:26.00 3rd Qu.:4.00
Max.   :33.00 Max.   :6.00
```

Figure 5 Descriptive statistics for age and family size

6. Categorical Descriptive Analysis

Frequency and Percentage are given for Gender, Marital Status, Occupation, Monthly Income, Educational Qualifications, Customer Type, Output and Feedback (Fernández-Cásseres and Russi-Pulgar, 2023).

The final sample consisted of 164 males, (57.54%) and 121 females (42.46%). For Customer Type, 162 customers (56.84%) were Regular, 105 (36.84%) were Frequent, and 18 (6.32%) were New.

Feedback was also mainly positive, with 231 observations (81.05%) marking positive and 54 (18.95%) marking negative feedback.

The distribution of occupation revealed that the largest group were the students, accounting for 144 observations (50.53%), followed by employees in terms of 94 observations (32.98%).

A considerable No Income category was also observed for monthly income with 131 observations (45.96%). These descriptive patterns offer background on the following statistical tests.

These descriptive patterns provide context for the subsequent statistical tests.

```
> # 12.3 Frequency and percentage analysis
> # -----
>
> categorical_variables <- c(
+   "Gender",
+   "Marital.Status",
+   "Occupation",
+   "Monthly.Income",
+   "Educational.Qualifications",
+   "Customer.Type",
+   "Output",
+   "Feedback"
+ )
>
> for (var in categorical_variables) {
+
+   cat("\n===== \n")
+   cat("Variable:", var, "\n")
+   cat("===== \n")
+
+   freq <- table(analysis_data[[var]])
+   percentage <- round(prop.table(freq) * 100, 2)
+
+   result <- data.frame(
+     Category = names(freq),
+     Frequency = as.vector(freq),
+     Percentage = as.vector(round(percentage, 2))
+   )
+
+   print(result)
+ }
```

=====			
variable: Educational.Qualifications			
=====			
	Category	Frequency	Percentage
1	Graduate	126	44.21
2	Ph.D	21	7.37
3	Post Graduate	125	43.86
4	School	11	3.86
5	Uneducated	2	0.70
=====			
variable: Customer.Type			
=====			
	Category	Frequency	Percentage
1	Frequent	105	36.84
2	New	18	6.32
3	Regular	162	56.84
=====			
variable: Output			
=====			
	Category	Frequency	Percentage
1	No	68	23.86
2	Yes	217	76.14
=====			
variable: Feedback			
=====			
	Category	Frequency	Percentage
1	Negative	54	18.95
2	Positive	231	81.05

Variable: Gender			
=====			
	Category	Frequency	Percentage
1	Female	121	42.46
2	Male	164	57.54
=====			
variable: Marital.Status			
=====			
	Category	Frequency	Percentage
1	Married	87	30.53
2	Prefer not to say	9	3.16
3	Single	189	66.32
=====			
variable: occupation			
=====			
	Category	Frequency	Percentage
1	Employee	94	32.98
2	House wife	9	3.16
3	Self Employed	38	13.33
4	Student	144	50.53
=====			
variable: Monthly.Income			
=====			
	Category	Frequency	Percentage
1	10001 to 25000	36	12.63
2	25001 to 50000	52	18.25
3	Below Rs.10000	19	6.67
4	More than 50000	47	16.49
5	No Income	131	45.96

Figure 6 Categorical frequency output

7. Exploratory Data Analysis

Five visualisations were produced to support the descriptive analysis.

Online Food Delivery Adoption

The Output distribution shows that the majority of observations fall within the **Yes** category. This indicates relatively high observed adoption within the analysed dataset.

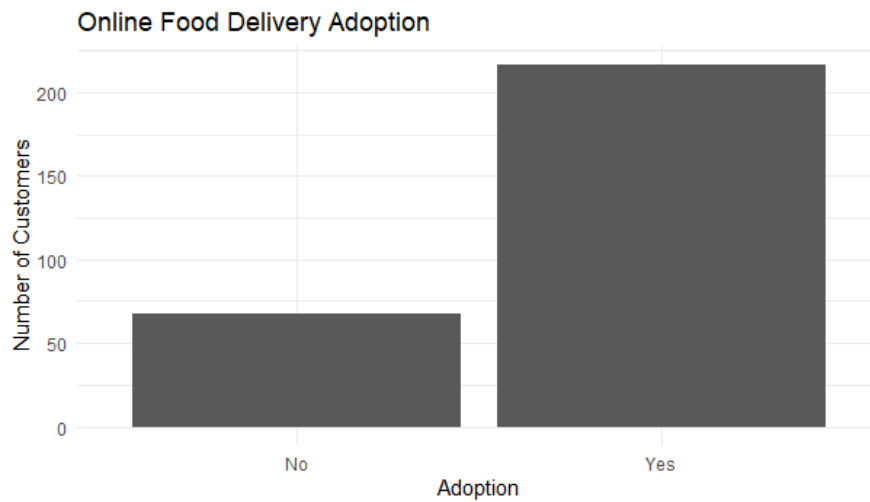


Figure 7 Online Food Delivery Adoption

Age Distribution

The age histogram shows the distribution of customer ages across the observed range of 18–33 years.

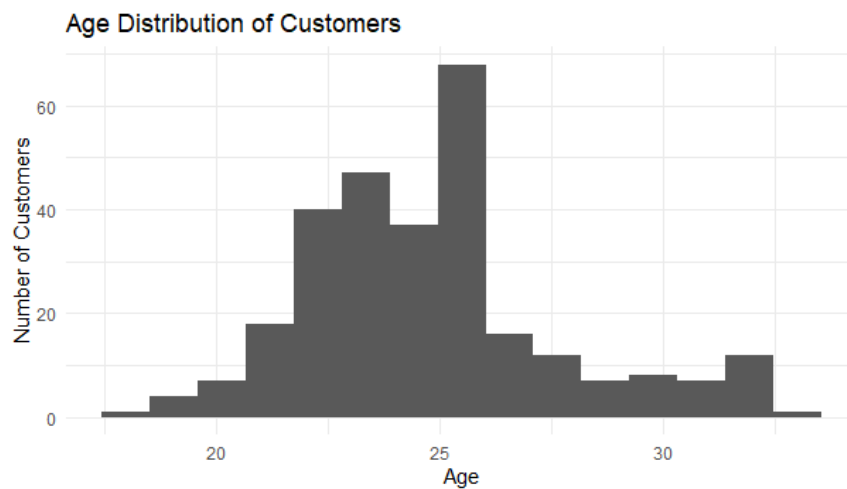


Figure 8 Age Distribution of Customers

Customer Type and Output

The 'Customer Type – Output' bar chart makes comparisons between Frequent, New and Regular customers.

The row percentages indicate that 75.24% of Frequent customers were adopted, 72.22% of New customers were adopted and 77.16% of Regular customers were adopted. These percentages are pretty close, but statistical tests will need to be performed to decide if there is a statistically significant difference.

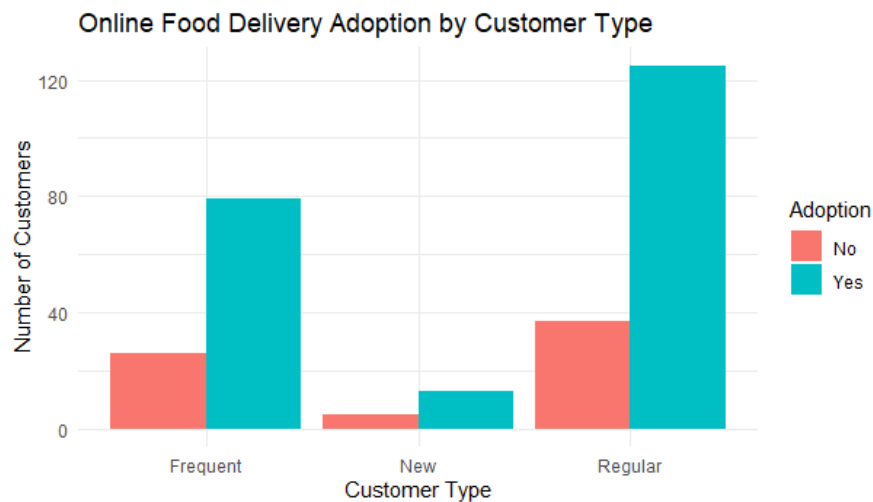


Figure 9 Online Food Delivery Adoption by Customer Type

Monthly Income and Output

The Monthly Income versus Output chart shows that there are variations in adoption rates between income groups.

For instance, the customers with No Income had 85.50% of customers who adopted while the customers with 25,001 – 50,000 resulted in 63.46% adoption. The difference patterns implied that income may possibly have a significant relationship to adoption as studied statistically in BQ2 (Chakraborty, Azam and Sana, 2022).

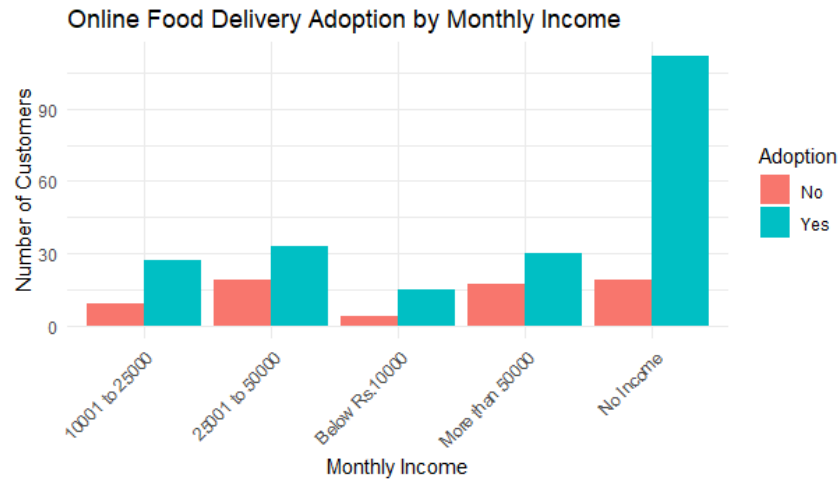


Figure 10 Online Food Delivery Adoption by Monthly Income

Occupation and Output

The data from the occupation analysis shows variations in uptake between occupational groups. Students adoption rate was 87.50%, and employees' adoption rate was 63.83% (Ariffin et al., 2021).

This visual difference will give a business rationale for deciding whether or not there is a statistical association between occupation and the risk of being adopted.

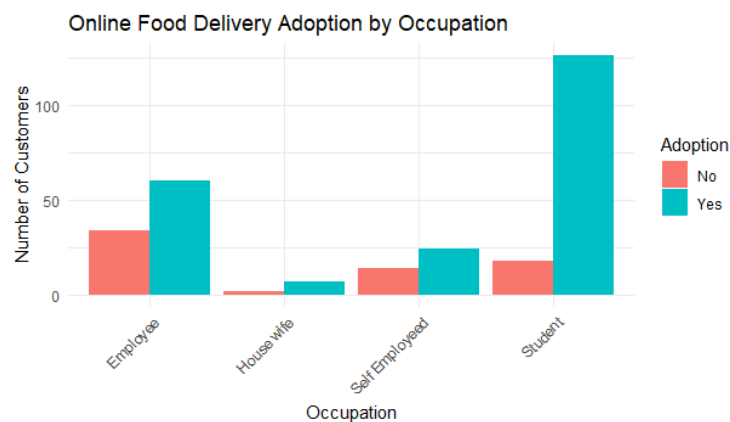


Figure 11 Online Food Delivery Adoption by Occupation

8. Inferential Statistical Analysis

8.1 Statistical Method

The three business questions involve relationships between categorical variables, so Pearson's chi-square test of independence was chosen (Deshmukh, 2024).

The test looks at if there are differences in the distribution of Output between the different levels of the explanatory variable.

The significance level was set at:

$$\alpha = 0.05$$

Therefore:

$$p < 0.05 \rightarrow \text{reject } H_0$$

$$p \geq 0.05 \rightarrow \text{fail to reject } H_0$$

Cramer's V was additionally calculated to assess the strength of the association.

9. Business Question 1: Customer Type and Output

The chi-square test for Customer Type and Output produced:

$$\chi^2 = 0.292$$

$$df = 2$$

$$p = 0.8642$$

$$\text{Cramer's V} = 0.0320$$

Due to the p value being much larger than the 0.05 level, the null hypothesis is not rejected.

Thus, no statistical association is seen between Customer Type and the uptake of online food delivery in this data set. It is a very slight association that is indicated by a significant Cramer's V of 0.032. This is in line with the relatively consistent descriptive row percentages for the type of customer.

Decision

Fail to reject H_0 .

```

>
> customer_chi <- chisq.test(
+   customer_output,
+   correct = FALSE
+ )

Warning message:
In stats::chisq.test(x, y, ...) :
  Chi-squared approximation may be incorrect

>
> customer_chi

      Pearson's Chi-squared test

data:  customer_output
X-squared = 0.29196, df = 2, p-value = 0.8642

> cat("\nChi-square statistic:",
+     as.numeric(customer_chi$statistic), "\n")

Chi-square statistic: 0.2919609
>
> cat("Degrees of freedom:",
+     as.numeric(customer_chi$parameter), "\n")
Degrees of freedom: 2
>
> cat("P-value:",
+     customer_chi$p.value, "\n")
P-value: 0.8641746

```

Figure 12 Pearson's Chi-Square test result for Customer Type and online food delivery adoption

```

> customer_chi_assumption <- chisq.test(
+   customer_output,
+   correct = FALSE
+ )

Warning message:
In stats::chisq.test(x, y, ...) :
  Chi-squared approximation may be incorrect

>
> cat("Expected frequencies:\n")
Expected frequencies:
> print(customer_chi_assumption$expected)

      No      Yes
Frequent 25.052632 79.94737
New      4.294737 13.70526
Regular  38.652632 123.34737

>
> cat("\nMinimum expected frequency:\n")

Minimum expected frequency:
> print(min(customer_chi_assumption$expected))
[1] 4.294737
>
> cat("\nNumber of expected frequencies below 5:\n")

Number of expected frequencies below 5:
> print(sum(customer_chi_assumption$expected < 5))
[1] 1

```

Figure 13 Chi-square assumption validation for Customer Type and online food delivery adoption

10. Business Question 2: Monthly Income and Output

The chi-square test for Monthly Income and Output produced:

$$\chi^2 = 14.942$$

$$df = 4$$

$$p = 0.004823$$

$$\text{Cramer's } V = 0.2290$$

The p-value obtained is less than 0.05, the level of significance. For this reason, the null hypothesis is turned down.

This will give statistical evidence of the relationship between the variables of interest: Monthly Income and Online Food Delivery Adoption (Gani et al., 2021). But the interpretation of statistical significance as a strong relationship is not warranted. This means that using the thresholds used in the interpretation, this Cramer's V of 0.229 reflects a low association.

The descriptive results give insight into the finding. The adoption rates were 63.46% for income between 25,001 and 50,000 and 85.50% for no income.

Decision

Reject H_0 .

```
> income_chi <- chisq.test(
+   income_output,
+   correct = FALSE
+ )

warning message:
In stats::chisq.test(x, y, ...) :
  Chi-squared approximation may be incorrect

>
> income_chi

      Pearson's Chi-squared test

data:  income_output
X-squared = 14.942, df = 4, p-value = 0.004823

> cat("\nChi-square statistic:",
+     as.numeric(income_chi$statistic), "\n")

Chi-square statistic: 14.94215

> cat("\nDegrees of freedom:",
+     as.numeric(income_chi$parameter), "\n")

Degrees of freedom: 4

>
> cat("\nP-value:",
+     income_chi$p.value, "\n")

P-value: 0.004822726

> # -----
```

Figure 14 Pearson's Chi-Square test result for Monthly Income and online food delivery adoption

11. Business Question 3: Occupation and Output

The chi-square test for Occupation and Output produced:

$$\chi^2 = 21.609$$

$$df = 3$$

$$p = 0.0000787$$

$$\text{Cramer's } V = 0.2754$$

The p-value is substantially below 0.05. Therefore, the null hypothesis is **rejected**.

The results showed statistical evidence of the relationship between occupation and adoption of online food delivery.

Cramer's $V = 0.275$ is a weak association. The descriptive analysis revealed that the students' adoption rate was 87.50%, which is higher compared to the employees' adoption rate of 63.83% and 63.16% for self-employed customers.

Thus, occupation seems to be statistically significantly correlated with adoption but the relationship is not strong.

Decision

Reject H_0 .

```

>
> occupation_chi <- chisq.test(
+   occupation_output,
+   correct = FALSE
+ )

warning message:
In stats::chisq.test(x, y, ...) :
  Chi-squared approximation may be incorrect

>
> occupation_chi

      Pearson's Chi-squared test

data:  occupation_output
X-squared = 21.609, df = 3, p-value = 7.867e-05

> cat("\nChi-square statistic:",
+   as.numeric(occupation_chi$statistic), "\n")

Chi-square statistic: 21.6089
>
> cat("Degrees of freedom:",
+   as.numeric(occupation_chi$parameter), "\n")
Degrees of freedom: 3
>
> cat("P-value:",
+   occupation_chi$p.value, "\n")
P-value: 7.86687e-05

```

Figure 15 Pearson's Chi-Square test result for Occupation and online food delivery adoption

12. Chi-Square Assumption Validation

Expected frequencies were examined before interpreting the chi-square results (Fernández-Cásseres and Russi-Pulgar, 2023).

For Customer Type, the minimum expected frequency was approximately **4.29**, with **1 of 6 cells (16.7%) below 5**.

For Monthly Income, the minimum expected frequency was approximately **4.53**, with **1 of 10 cells (10%) below 5**.

For Occupation, the minimum expected frequency was approximately **2.15**, with **1 of 8 cells (12.5%) below 5**.

The analysis therefore identified some expected frequencies below 5, and R produced the warning: “Chi-squared approximation may be incorrect.”

Due to this warning, the regular Pearson results were not interpreted independently from one another. To test the robustness a Monte Carlo simulation was carried out with 10,000 replications.

```

> customer_chi_assumption <- chisq.test(
+   customer_output,
+   correct = FALSE
+ )

warning message:
In stats::chisq.test(x, y, ...) :
  Chi-squared approximation may be incorrect

>
> cat("Expected frequencies:\n")
Expected frequencies:
> print(customer_chi_assumption$expected)

           No      Yes
Frequent 25.052632 79.94737
New      4.294737 13.70526
Regular 38.652632 123.34737

>
> cat("\nMinimum expected frequency:\n")

Minimum expected frequency:
> print(min(customer_chi_assumption$expected))
[1] 4.294737

>
> cat("\nNumber of expected frequencies below 5:\n")

Number of expected frequencies below 5:
> print(sum(customer_chi_assumption$expected < 5))
[1] 1

```

Figure 16 Chi-square assumption validation for Customer Type and online food delivery adoption

13. Monte Carlo Robustness Check

The Monte Carlo chi-square procedure produced the following results:

Business Question	Pearson p-value	Monte Carlo p-value
Customer Type and Output	0.86417	0.87621
Monthly Income and Output	0.004823	0.005000
Occupation and Output	0.0000787	0.00009999

The robustness results support the original conclusions.

The p-value for Customer Type is greater than 0.05 for both, so their null hypothesis is not rejected.

For Monthly income, the p values are both less than 0.05 which means that the null hypothesis is again rejected. The null hypothesis is rejected for both p-values in Occupation which are still well below 0.05.

Thus, the key findings of the statistics remain robust with respect to Monte Carlo robustness check.

14. Overall Findings and Business Implications

The analysis gives the different answers to the three business questions. First, the relationship of Customer Type to Online Food Delivery Adoption was not statistically significant. The Cramer's V is very close to zero, indicating that the differences between the Frequent, New and Regular customers were quite minute in reality.

Second, Monthly income was found to be statistically significant in terms of the adoption. But it is important to note that the low value of the Cramer's V does not suggest that income is a very strong factor when deciding to adopt. Third, the effect size of Occupation was statistically small but it was related to adoption. Students had higher adoption rates than some other occupational categories, especially.

These results indicate that from a business point of view, a business dealing with customers shouldn't solely segment the customers based on Customer Type (Chiu, Badu-Baiden and Cho, 2024). Rather, the ways in which people utilize their occupations and income might be part of segmentation strategies. For instance, student-focused marketing promotions and flexible offers may be incorporated in marketing campaigns, while income-based rather than assuming income determine food delivery behavior online. But the statistical correlations must not be taken to mean that there is cause and effect. The analysis does not show that there is a direct relationship between income and/or occupation and customers' online food order and delivery.

15. Limitations

There are some drawbacks to be taken into account. First of all, the data itself provides observational customer information, which means that the fact that we found association, doesn't imply that any of the factors studied is a cause or effect.

Secondly, there is a significant amount of the data in the original set which are repeated. While exact duplicates were eliminated, the source of the duplicates is not determinable from the data set.

Thirdly, some categories contain low numbers of frequencies. This was a factor in the chi-square approximation warning and led to a Monte Carlo robustness check. Lastly, only three selected

explanatory variables are selected for analysis. Other factors like location, gender, age, education and feedback also can play a role in the adoption of online food delivery.

16. Recommendations and Future Work

According to results, companies may be inspired to consider occupation and income in the creation of customer segmentation methods (Shankar et al., 2024). Observed adoption rate of students was high and students are an important customer group. Student-targeted promotions, student-friendly meal packages, and targeted digital campaigns, for example, might be considered.

The associations were, however, weak and not strong and therefore, businesses should not rely on one variable alone to identify customers. Further studies could use more complex statistical modelling to explore the interactions among a set of customer attributes. For instance, logistic regression could simulate the chance of adoption taking multiple demographic and socioeconomic factors into account.

The relationships could also be explored with larger and more representative data sets in the future.

17. Conclusion

This Project analyzed Online Food Delivery Adoption with an Online Food Delivery Dataset using statistical analysis. Data was checked, cleaned and validated prior to analysis. Following the process of removing redundant information and duplicate observations, the final data-set consisted of 285 distinct observations and 13 variables that did not have any missing items.

Using descriptive statistics and EDA, differences were found in the adoption behavior of customer groups. Pearson's chi-square tests revealed that there was no significant association between Customer Type and Output, but there was a significant association between Monthly Income and Occupation and Output. The effect sizes shown suggested a negligible association with Customer Type, and weak associations with Monthly Income and Occupation. Most importantly, the hypothesis decisions for both the Monte-Carlo robustness analysis and the Pearson tests were the same.

In general, their results show that there are relationships between Monthly Income, Occupation, and the adoption of online food deliveries in this data set, but not for Customer Type. The result of these findings can contribute to evidence-based customer segmentation, and the constraints suggest there is scope to enrich future decision making by analyzing the data in greater depth with a larger set and applying multivariable techniques.

References

- Ahmad, A. (2023). Consumer adoption of online food delivery services: Scale development and validation. *Journal of Foodservice Business Research*, 28(2), pp.277–294. doi:10.1080/15378020.2023.2244669.
- Ariffin, S., Abdul Manan, H., Ahmad, N., Muhammad, N.S., Hamdan, F. and S Kelana, N.S. (2021). Continuous intention to use technology of online food delivery services among young adults / Shahira Ariffin ... [et al.]. *Advances in Business Research International Journal*, [online] 7(1), pp.56–64. Available at: <https://ir.uitm.edu.my/id/eprint/49505/> [Accessed 15 Sept. 2026].
- Aslam, W., Ham, M. and Arif, I. (2021). Technology at the dining table: Linking perceived value, service recovery, and continuous intention to use food delivery applications. *Revista Brasileira de Gestão de Negócios*, [online] 23(4), pp.600–618. doi:10.7819/rbgn.v23i4.4135.
- Bao, Z. and Zhu, Y. (2021). Why customers have the intention to reuse food delivery apps: Evidence from China. *British Food Journal*. doi:10.1108/bfj-03-2021-0205.
- Chakraborty, S., Azam, M.K. and Sana (2022). Factors affecting the behavioural intentions of Indian millennials. *International Journal of Online Marketing*, 12(1), pp.1–16. doi:10.4018/ijom.306975.
- Chiu, W., Badu-Baiden, F. and Cho, H. (2024). Consumers' intention to use online food delivery services: A meta-analytic structural equation modeling approach. *International Journal of Consumer Studies*, 48(3). doi:10.1111/ijcs.13052.
- Chotigo, J. and Kadono, Y. (2021). Comparative analysis of key factors encouraging food delivery app adoption before and during the COVID-19 pandemic in Thailand. *Sustainability*, 13(8), p.4088. doi:10.3390/su13084088.
- Deshmukh, P. (2024). Chi-square test: A parametric or nonparametric enigma in public health research. *Indian Journal of Community and Family Medicine*, 10(2), pp.71–73. doi:10.4103/ijcfm.ijcfm_112_24.
- Fernández-Cásseres, M.A. and Russi-Pulgar, D. (2023). *Chapter 30 - categorical variable analyses: Chi-square, Fisher exact, mantel–haenszel*. [online] ScienceDirect. Available at:

<https://www.sciencedirect.com/science/article/pii/B9780323912594000436> [Accessed 15 Sept. 2026].

Francioni, B., Curina, I., Hegner, S.M. and Cioppi, M. (2022). Predictors of continuance intention of online food delivery services: Gender as moderator. *International Journal of Retail & Distribution Management*, 50(12). doi:10.1108/ijrdm-11-2021-0537.

Gani, M.O., Faroque, A.R., Muzareba, A.M., Amin, S. and Rahman, M. (2021). An integrated model to decipher online food delivery app adoption behavior in the COVID-19 pandemic. *Journal of Foodservice Business Research*, 26(2), pp.123–163. doi:10.1080/15378020.2021.2006040.

Hong, C., Choi, H.H., Choi, E.-K.C. and Joung, H.-W.D. (2021). Factors affecting customer intention to use online food delivery services before and during the COVID-19 pandemic. *Journal of Hospitality and Tourism Management*, [online] 48(1), pp.509–518. doi:10.1016/j.jhtm.2021.08.012.

Hu, L.-Y.R., He, Y., Irimata, K.E. and Beresovsky, V. (2025). Much ado about survey tables: A comparison of chi-square tests and software to analyze categorical survey data. *The American Statistician*, 79(4), pp.480–491. doi:10.1080/00031305.2025.2501800.

Osei, I., Carie, C.A., Anamalamudi, S. and Boadu, D.O. (2026). Advancing human-computer interaction research through structural equation modelling. *Frontiers in Human Dynamics*, [online] 8. doi:10.3389/fhumd.2026.1741286.

Raza, A., Asif, M. and Akram, M. (2022). Give your hunger a new option: Understanding consumers’ continuous intention to use online food delivery apps using trust transfer theory. *International Journal of Consumer Studies*, 47(2). doi:10.1111/ijcs.12845.

Shankar, A., Jebarajakirthy, C., Haroon Iqbal Maseeh, Preeti Nayal, Kumar, A. and Krishnan, C. (2024). Why do consumers choose online food delivery services? A meta-analytic review. *International Journal of Hospitality Management*, [online] 123, pp.103921–103921. doi:10.1016/j.ijhm.2024.103921.

Sihler, F. and Tichy, M. (2025). Statically analyzing the dataflow of R programs. *Proceedings of the ACM on Programming Languages*, 9(OOPSLA2), pp.1034–1062. doi:10.1145/3763087.

Tierney, N. and Cook, D. (2023). Expanding tidy data principles to facilitate missing data exploration, visualization and assessment of imputations. *Journal of Statistical Software*, 105(7). doi:10.18637/jss.v105.i07.